



(12) **United States Patent**
Olvera et al.

(10) **Patent No.:** **US 12,409,536 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

- (54) **TOOL FOR POWERED SPRAYER**
- (71) Applicant: **TECHTRONIC CORDLESS GP**,
Anderson, SC (US)
- (72) Inventors: **Eduardo Olvera**, Pendleton, SC (US);
Graham M. Schaafsma, Raleigh, NC
(US)
- (73) Assignee: **Techtronic Cordless GP**, Anderson, SC
(US)
- (*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

- (21) Appl. No.: **18/671,063**
- (22) Filed: **May 22, 2024**

- (65) **Prior Publication Data**
US 2024/0308041 A1 Sep. 19, 2024

- Related U.S. Application Data**
- (63) Continuation of application No. 17/552,529, filed on
Dec. 16, 2021, now Pat. No. 12,017,331.
- (60) Provisional application No. 63/134,386, filed on Jan.
6, 2021.
- (51) **Int. Cl.**
B25B 27/14 (2006.01)
B05B 15/60 (2018.01)
- (52) **U.S. Cl.**
CPC **B25B 27/14** (2013.01); **B05B 15/60**
(2018.02)
- (58) **Field of Classification Search**
CPC B25F 1/02
USPC 81/177.4, 490
See application file for complete search history.

- (56) **References Cited**
- U.S. PATENT DOCUMENTS
- | | | | |
|---------------|---------|-----------------|------------------------|
| 4,219,161 A | 8/1980 | Freissle | |
| 4,302,990 A * | 12/1981 | Chrichton | B25G 1/085
81/177.4 |
| 4,434,828 A * | 3/1984 | Trincia | B25G 1/085
81/177.4 |
| 4,533,005 A | 8/1985 | Morris | |
| 4,583,694 A | 4/1986 | Williams et al. | |
| 5,337,958 A | 8/1994 | Hennessy et al. | |
| 5,421,522 A | 6/1995 | Bowen | |
- (Continued)

FOREIGN PATENT DOCUMENTS

- | | | |
|----|-------------|--------|
| CN | 104289349 B | 8/2016 |
| DE | 20203740 U1 | 5/2002 |
- (Continued)

OTHER PUBLICATIONS

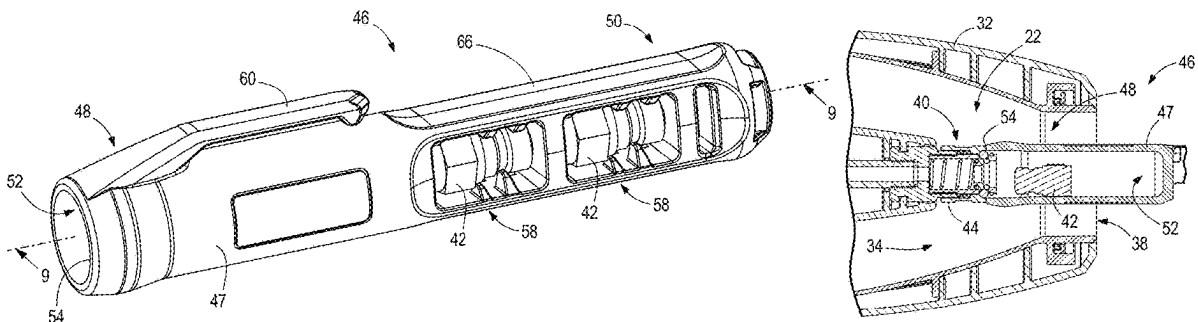
Extended European Search Report for Application No. 21217273.8
dated Jun. 2, 2022 (8 pages).

Primary Examiner — Jason L Vaughan

(74) *Attorney, Agent, or Firm* — Michael Best &
Friedrich LLP

- (57) **ABSTRACT**
- The present disclosure is directed to a tool for operating a
nozzle holder. The tool may include an elongated body
extending between a first end and a second end opposite the
first end, the second end defining a recess configured to
selectively receive and hold a portion of the spray nozzle.
The tool may include a nozzle storage receptacle formed in
the elongated body between the first end and the second end,
the nozzle storage receptacle configured to selectively
receive and hold the spray nozzle. The nozzle storage
receptacle defines an aperture that extends through the
elongated body between a first lateral side of the elongated
body and a second lateral side of the elongated body
opposite the first lateral side.

20 Claims, 8 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

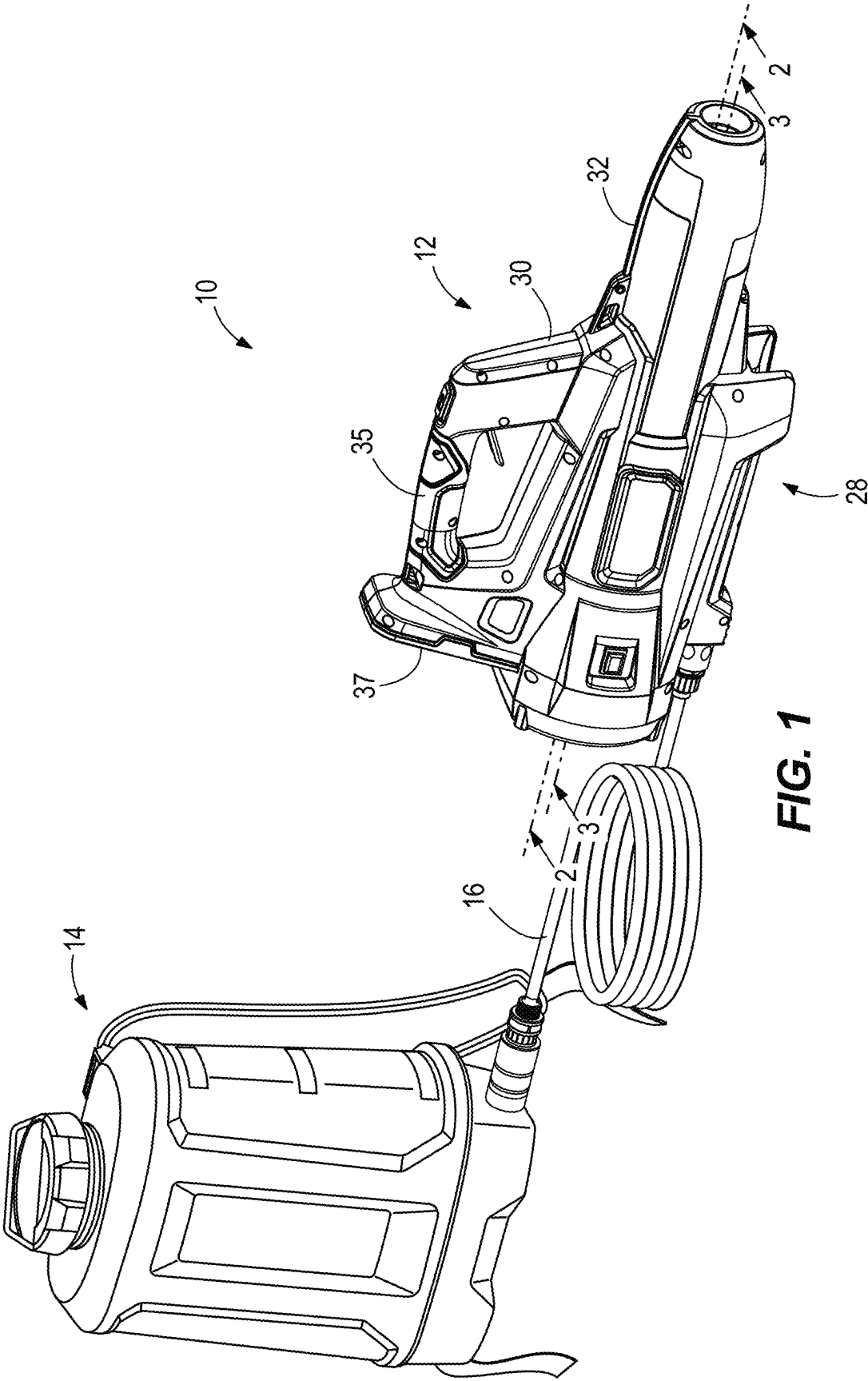
5,479,670	A	1/1996	Coffey	
6,164,172	A *	12/2000	Huang	B25G 1/085 81/439
6,216,959	B1	4/2001	Garrison et al.	
6,234,411	B1	5/2001	Walker et al.	
6,739,224	B1 *	5/2004	Wershe	B25G 1/085 81/177.4
6,969,057	B2	11/2005	Baltz et al.	
7,458,524	B2	12/2008	Prieto et al.	
7,648,086	B2	1/2010	Struck	
7,770,815	B2	8/2010	Green et al.	
8,308,083	B2	11/2012	Woodgate et al.	
8,590,810	B2	11/2013	Woodgate et al.	
8,672,243	B2	3/2014	Turcic et al.	
9,283,574	B2	3/2016	Leistiko	
9,303,393	B2	4/2016	Bush	
10,035,161	B2	7/2018	Robert	
10,087,609	B2	10/2018	Bush	
10,137,468	B2	11/2018	Robert	
10,322,424	B2	6/2019	Wright	
10,378,192	B2	8/2019	Bush	
10,434,533	B2	10/2019	Robert	

10,589,298	B2	3/2020	Wright	
2003/0230643	A1 *	12/2003	Bugarin	B05B 1/16 239/390
2007/0257138	A1	11/2007	Riese	
2010/0270337	A1	10/2010	Green et al.	
2010/0327086	A1	12/2010	Hassler	
2011/0088522	A1 *	4/2011	Floyd	B25B 15/00 81/177.85
2012/0031993	A1	2/2012	Leistiko	
2013/0245576	A1	9/2013	Hoogenakker et al.	
2013/0327850	A1	12/2013	Joseph et al.	
2016/0221160	A1	8/2016	Boland et al.	
2017/0173607	A1	6/2017	Wright	
2018/0178226	A1	6/2018	Morgan et al.	
2019/0060922	A1	2/2019	Wright	
2020/0023384	A1	1/2020	Ketterer	
2020/0121867	A1	4/2020	Wright	
2020/0238308	A1	7/2020	Forgey et al.	
2020/0384616	A1	12/2020	Anex	

FOREIGN PATENT DOCUMENTS

WO	2015051385	A1	4/2015
WO	2020005691	A1	1/2020

* cited by examiner



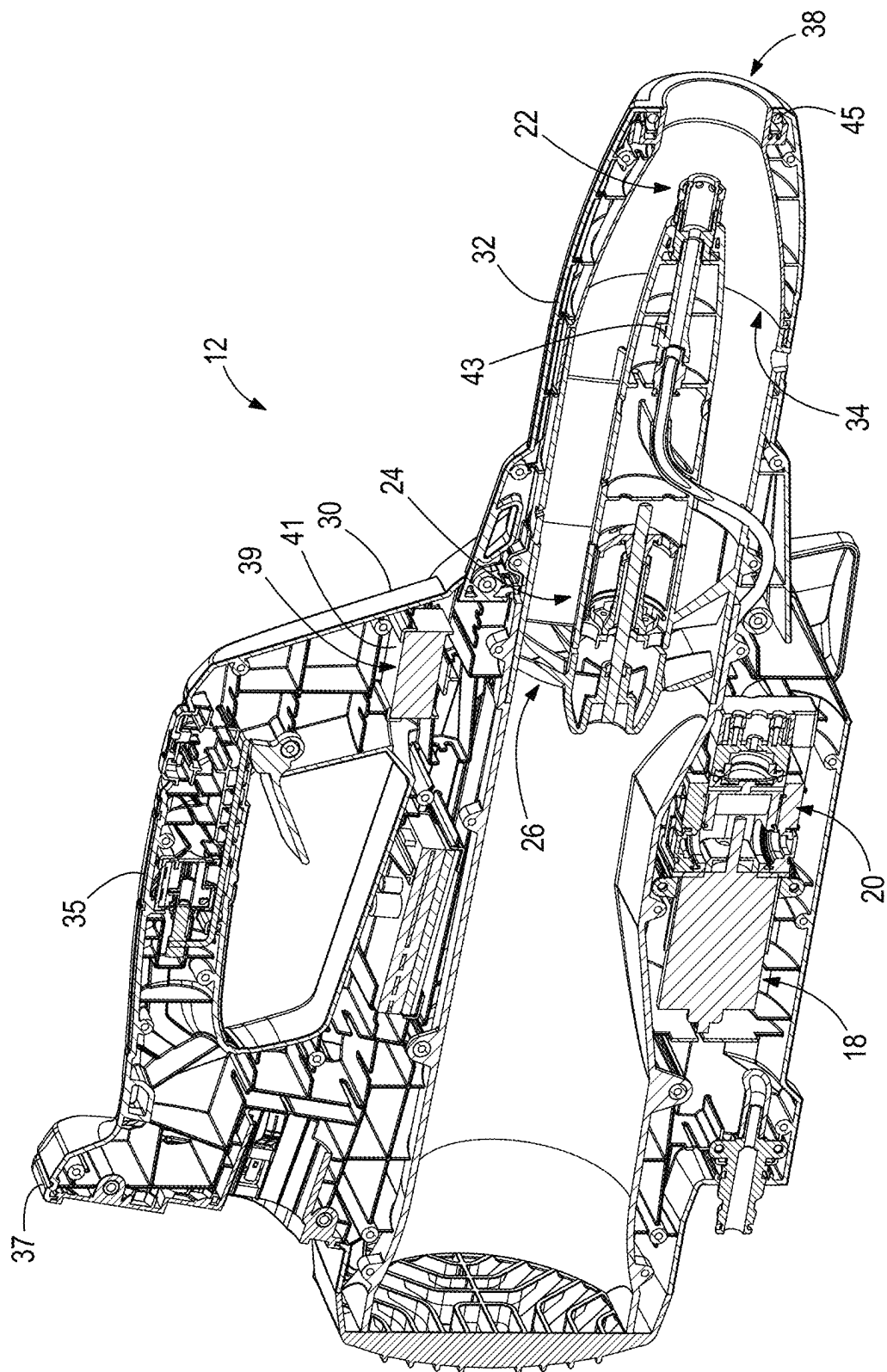


FIG. 2

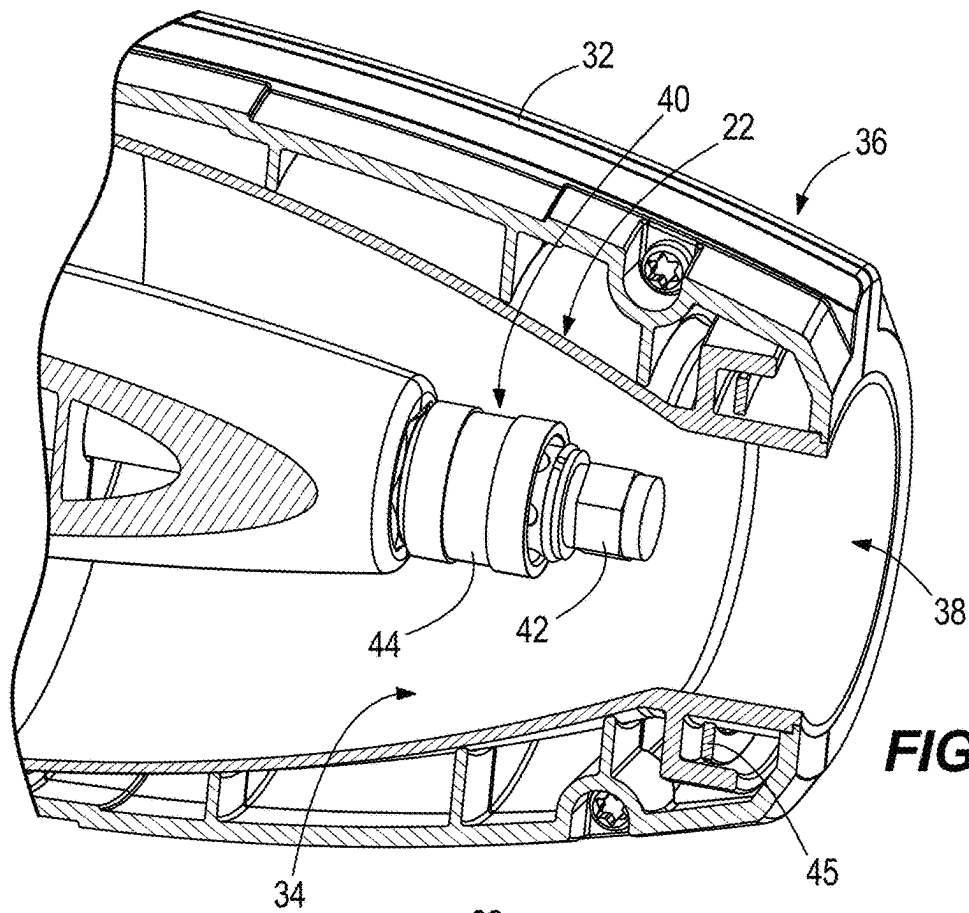


FIG. 3

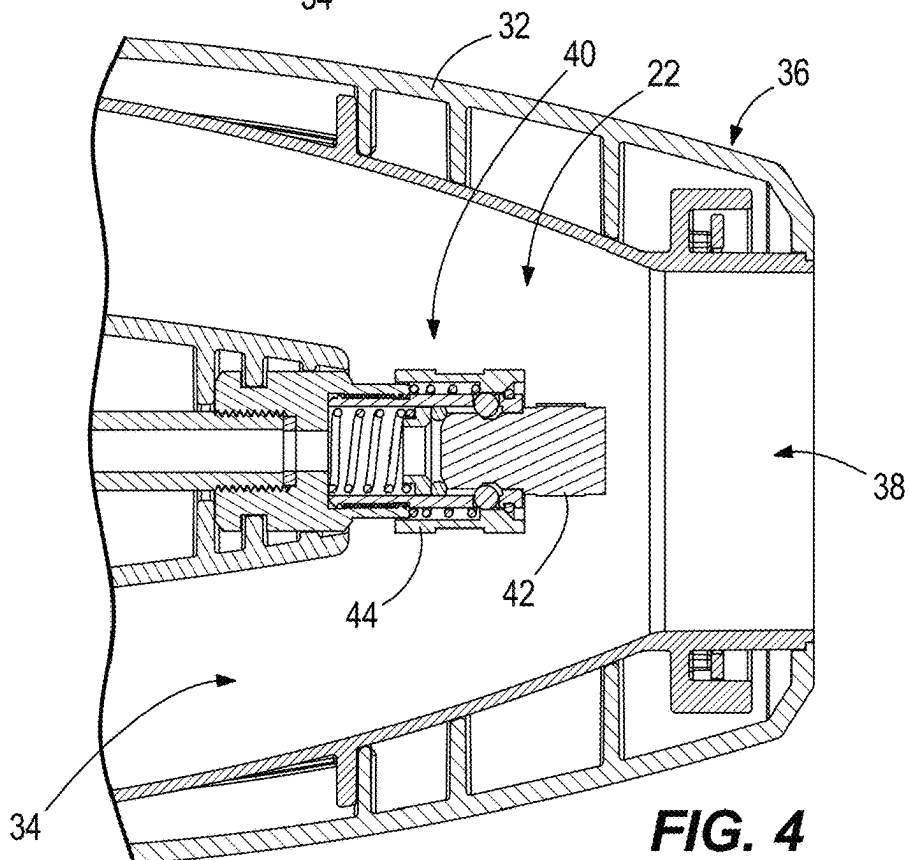
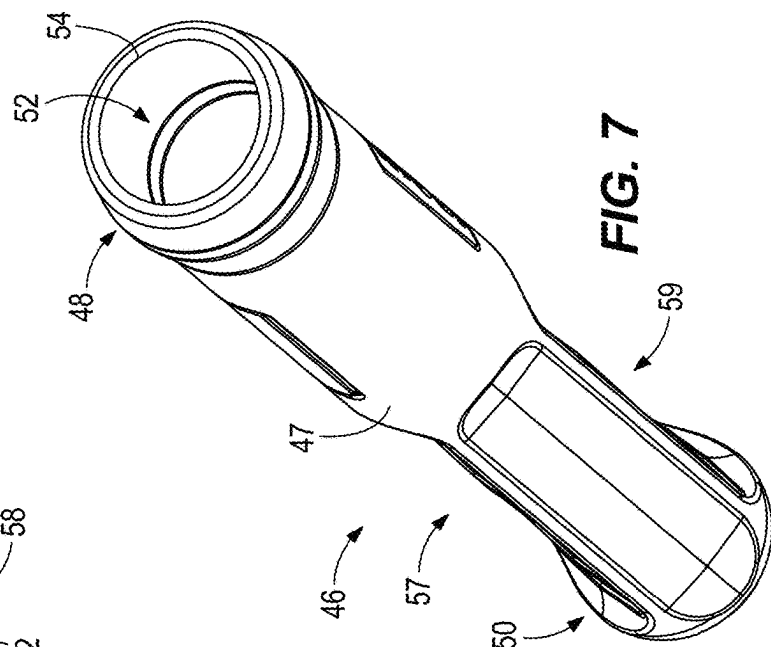
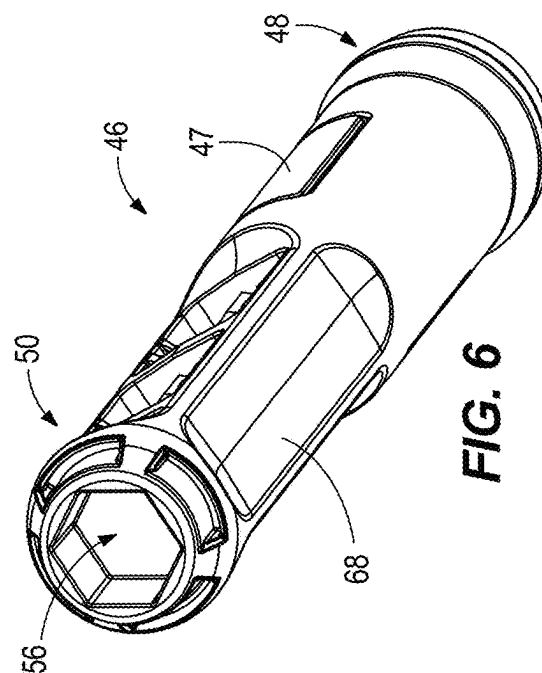
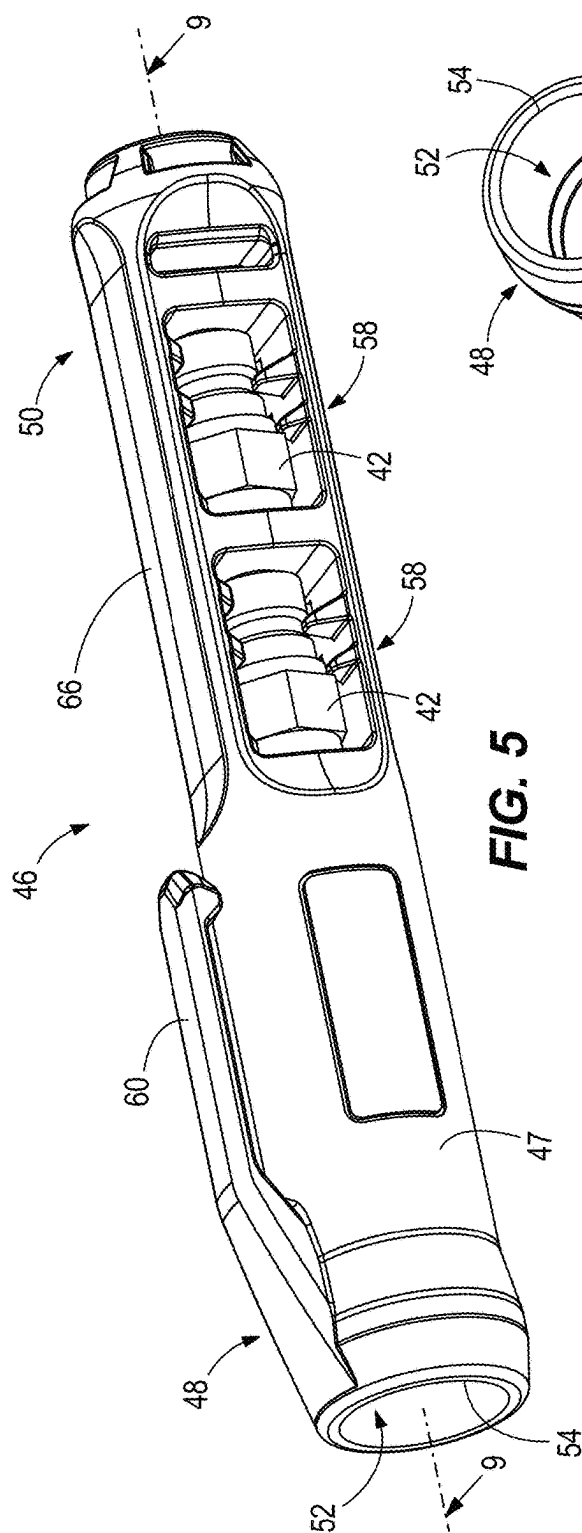


FIG. 4



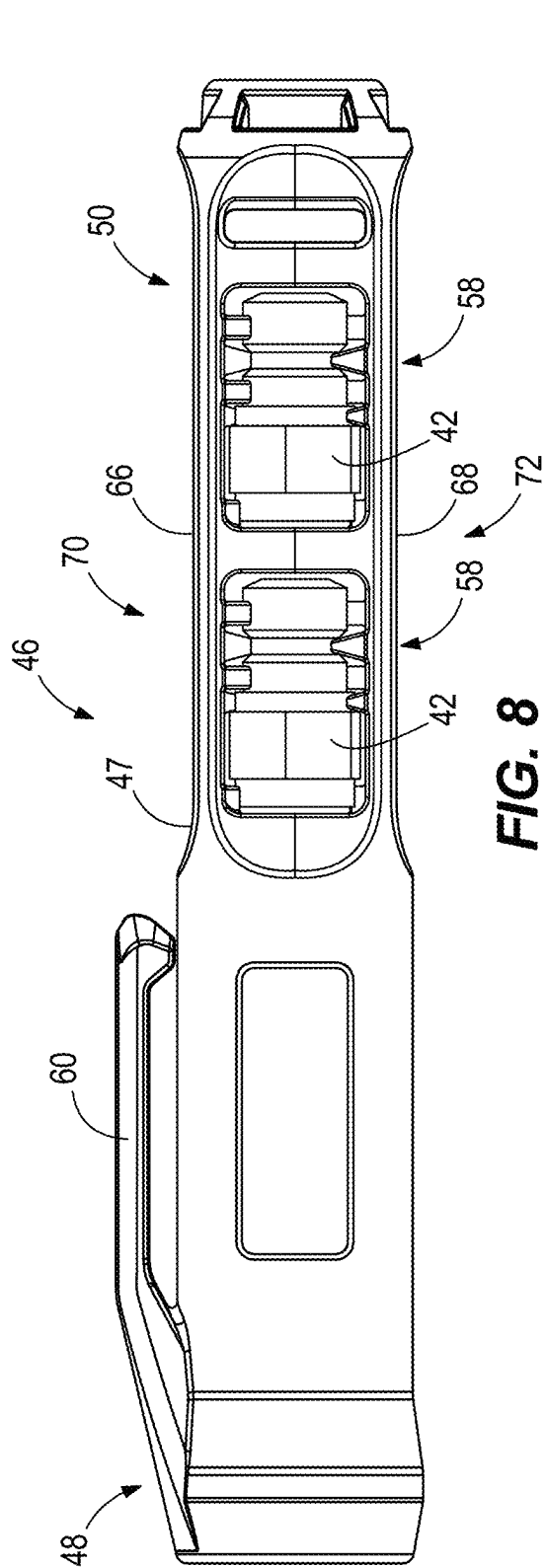


FIG. 8

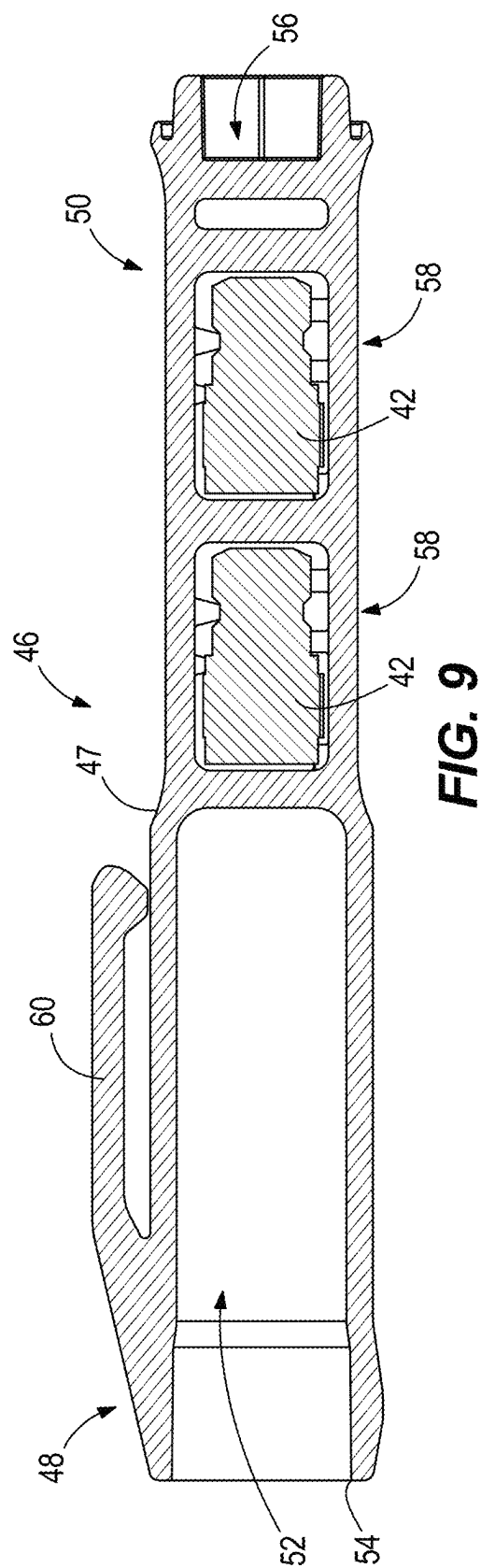


FIG. 9

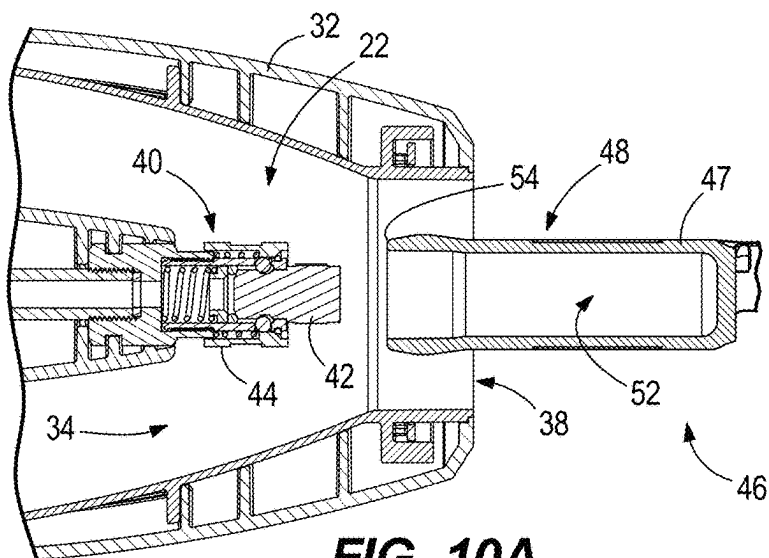


FIG. 10A

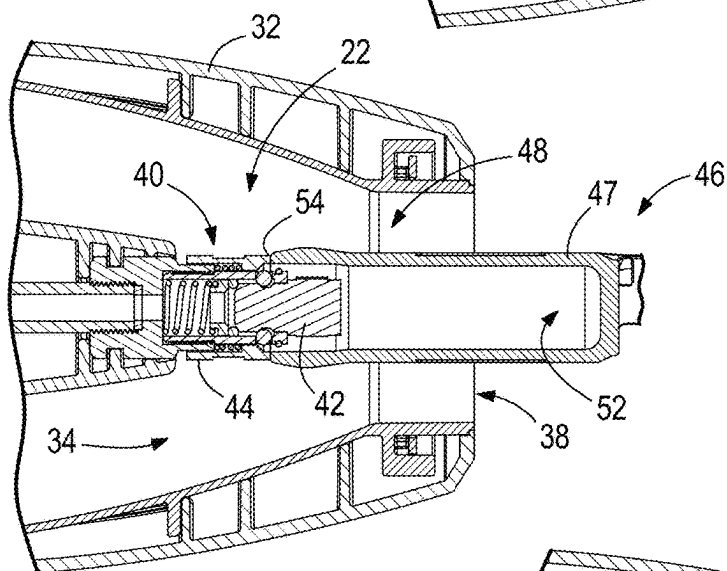


FIG. 10B

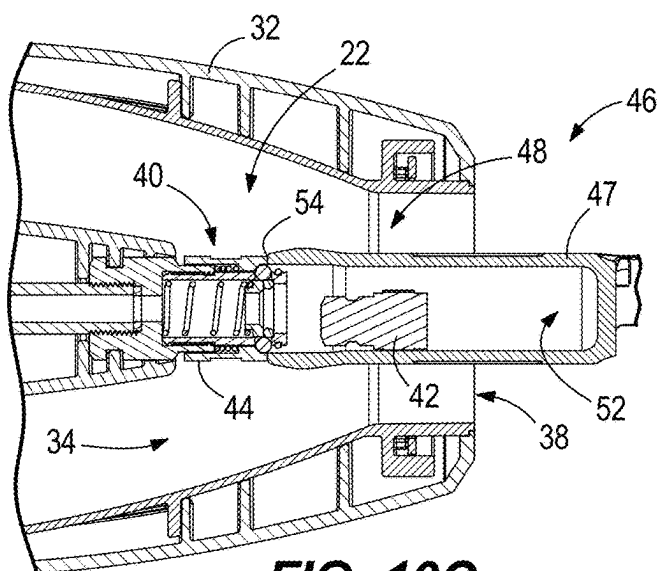
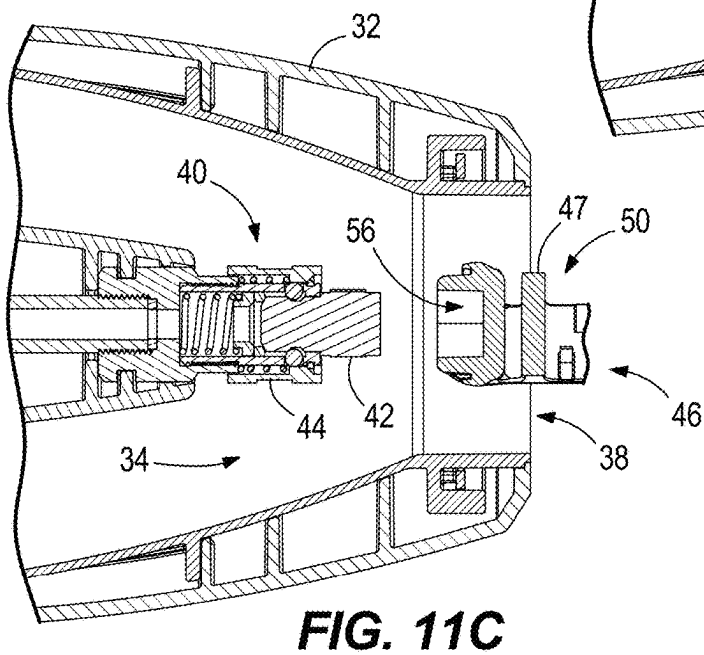
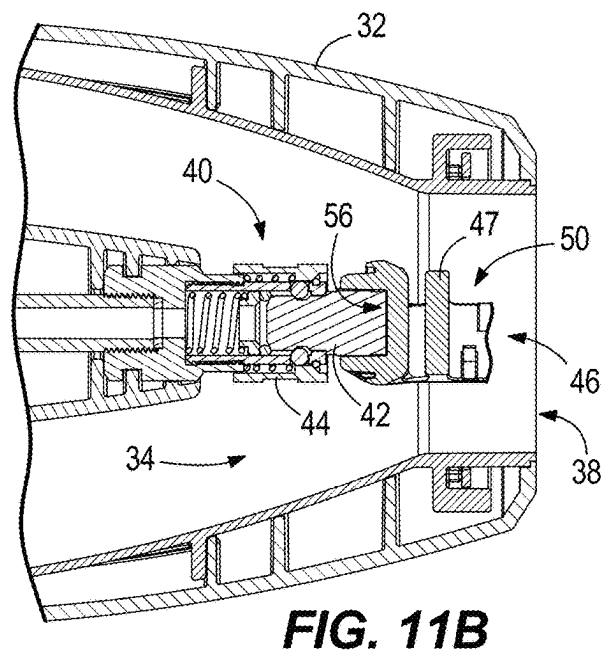
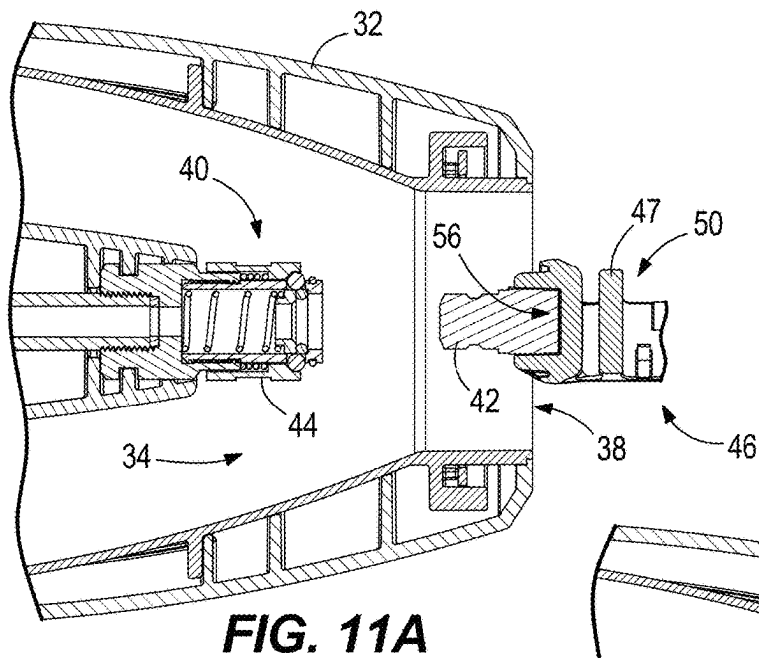


FIG. 10C



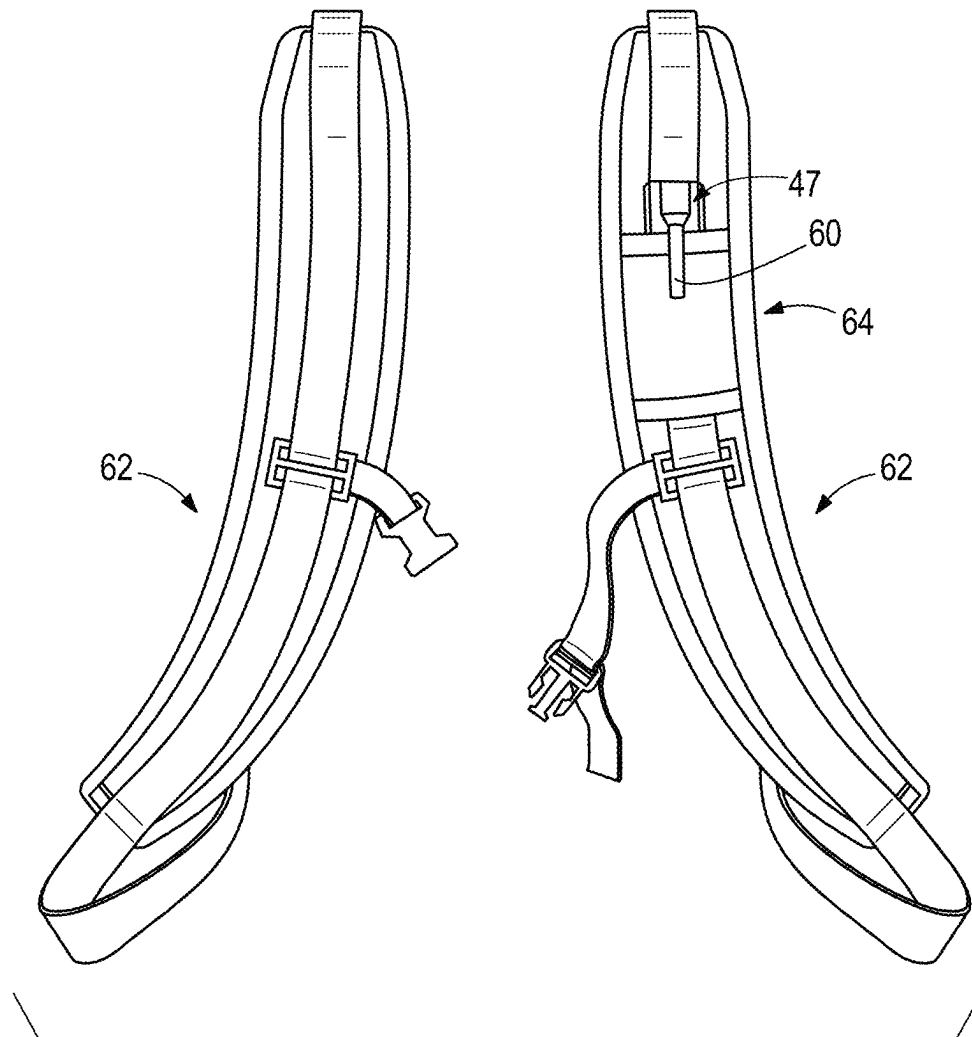


FIG. 12

1

TOOL FOR POWERED SPRAYER**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a continuation of U.S. patent application Ser. No. 17/552,529, filed Dec. 16, 2021, which claims priority to U.S. Provisional Patent Application No. 63/134,386, filed Jan. 6, 2021, the entire contents of all of which are incorporated herein by reference.

FIELD OF THE DISCLOSURE

The present invention relates to powered sprayers, and more particularly to tools for operating a nozzle of a powered sprayer.

BACKGROUND OF THE DISCLOSURE

A powered sprayer can include a removable nozzle. A tool can facilitate removing or installing the nozzle in the powered sprayer.

SUMMARY OF THE DISCLOSURE

The present disclosure provides, in one aspect, a tool for operating a nozzle holder, the nozzle holder configured to selectively and removably couple a spray nozzle to a spray device, the tool including: an elongated body extending between a first end and a second end opposite the first end, the second end defining a recess configured to selectively receive and hold a portion of the spray nozzle; and a nozzle storage receptacle formed in the elongated body between the first end and the second end, the nozzle storage receptacle configured to selectively receive and hold the spray nozzle; wherein the nozzle storage receptacle defines an aperture that extends through the elongated body between a first lateral side of the elongated body and a second lateral side of the elongated body opposite the first lateral side.

The present disclosure provides, in another aspect, a tool for operating a nozzle holder, the nozzle holder configured to selectively and removably couple a spray nozzle to a spray device, the tool including: an elongated body extending between a first end and a second end opposite the first end, the second end defining a recess configured to selectively receive and hold a portion of the spray nozzle; and a nozzle storage receptacle formed in the elongated body between the first end and the second end, the nozzle storage receptacle configured to selectively hold and receive the spray nozzle inserted from a first lateral side of the elongated body and an opposite, second lateral side of the elongated body.

The present disclosure provides, in another aspect, a tool for operating a nozzle holder, the nozzle holder configured to selectively and removably couple a spray nozzle to a spray device, the tool including: an elongated body extending between a first end and a second end; an annular rim defined by the first end, the annular rim being configured to selectively engage the nozzle holder to release the spray nozzle; and a nozzle storage receptacle formed in the elongated body between the first end and the second end, the nozzle storage receptacle configured to selectively hold and receive the spray nozzle inserted from a first lateral side of the elongated body and an opposite, second lateral side of the elongated body.

2

Other aspects of the disclosure will become apparent by consideration of the detailed description and accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a fogger according to an embodiment of the disclosure.

FIG. 2 is a perspective cross-sectional view of a sprayer unit of the fogger of FIG. 1, taken along line 2-2 of FIG. 1.

FIG. 3 is a perspective cross-sectional view of portions of the sprayer unit of FIG. 2, taken along line 3-3 of FIG. 1.

FIG. 4 is a cross-sectional view of portions of the sprayer unit of FIG. 2, taken along line 2-2 of FIG. 1.

FIGS. 5-7 are perspective views of a tool for operating a nozzle holder of the fogger of FIG. 1.

FIG. 8 is a side view of the tool of FIG. 5.

FIG. 9 is a cross-sectional view of the tool of FIG. 5, taken along line 9-9 of FIG. 5.

FIGS. 10A-10C are cross-sectional views of portions of the sprayer unit of FIG. 2 and of the tool of FIG. 5, illustrating a process for removing a spray nozzle from the nozzle holder using the tool.

FIGS. 11A-11C are cross-sectional views of portions of the sprayer unit of FIG. 2 and of the tool of FIG. 5, illustrating a process for installing the spray nozzle into the nozzle holder using the tool.

FIG. 12 illustrates a pair of backpack straps that can be attached to a backpack for carrying a remote reservoir of the fogger of FIG. 1.

Before any embodiments of the application are explained in detail, it is to be understood that the application is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the following drawings. The application is capable of other embodiments and of being practiced or of being carried out in various ways.

DETAILED DESCRIPTION

FIGS. 1 and 2 illustrate a spray device configured as an electrostatic powered sprayer or fogger 10 for spraying a fine mist of atomized liquid disinfecting solution to disinfect surfaces. The fogger 10 includes a handheld sprayer unit 12 fluidly coupled to a remote reservoir 14 via a hose 16. The sprayer unit 12 includes a first motor 18 that drives a liquid pump 20 to draw the liquid solution from the reservoir 14 and direct the solution through an atomizing nozzle assembly 22. The sprayer unit 12 also includes a second motor 24 that drives a blower 26 to direct an airflow past the nozzle assembly 22 to increase the throw of the atomized solution.

The sprayer unit 12 further includes a housing 28 that includes a main housing 30 in which the motor 18 and pump 20 are supported, and a tube 32 that defines an airflow channel 34. The main housing 30 defines a handle portion 35 that may be grasped by the user during use to manipulate the orientation and position of the sprayer unit 12. The main housing 30 also defines a battery receptacle 37 that selectively couples to a removable and rechargeable battery pack (not shown) to provide DC electrical power to the sprayer 10. In alternative embodiments, the sprayer 10 may be AC powered (e.g., plugged into a standard home electrical socket).

With reference to FIGS. 2 and 3, the sprayer 10 also includes an electrostatic charging circuit 39 operable to impart an electrostatic charge in the atomized liquid droplets exiting the nozzle assembly 22 during operation of the

3

sprayer 10. The electrostatic charging circuit 39 includes a high voltage supply module 41 supported within the main housing 30, a first or high voltage electrode assembly 43 positioned in-line between the pump 20 and the nozzle assembly 22, and a second electrode assembly or grounding ring assembly 45 suspended in front of the nozzle assembly 22. The high voltage supply module 41 receives power from the battery pack and supplies power to the high voltage electrode assembly 43. The grounding ring assembly 45 is electrically coupled to the high voltage supply module 41 to complete the charging circuit 39. During operation of the sprayer 10, the liquid solution passes through the high voltage electrode assembly 43 and is directly charged by direct contact therewith. An electric field develops between the grounding ring assembly 45 and any conductive material (such as, e.g., the liquid solution) in electrical contact with the high voltage electrode assembly 43. As the atomized spray droplets of the liquid solution exit the nozzle assembly 22, the droplets pass through the interior of the grounding ring assembly 45 and through the electric field, thereby acquiring an electrostatic charge. The acquired charge enables the spray droplets to better adhere or "stick" to targeted surfaces.

The nozzle assembly 22 is suspended within the tube 32 proximate a forward end 36 that defines an outlet opening 38. The nozzle assembly 22 includes a quick-connect nozzle holder 40 and a removable, replaceable nozzle 42. The nozzle 42 can be removed from the nozzle holder 40 (e.g., for cleaning) and can also be replaced with a different nozzle (e.g., having different spray characteristics). Specifically, the nozzle holder 40 includes a retractable outer ring 44 that is movable between a holding position (FIG. 4), in which the nozzle 42 is securely held within the nozzle holder 40, and a release position (FIGS. 10B and 10C), in which the nozzle 42 is released and ejected from the nozzle holder 40. In other embodiments (not shown), the nozzle holder 40 can receive and hold the nozzle by other means. For example, the nozzle holder 40 can include internal threads for threadably receiving an externally threaded nozzle. In another example, the nozzle holder 40 can receive the nozzle in a bayonet lock arrangement.

Since the nozzle assembly 22 resides within the tube 32, it must be accessed through the outlet opening 38 to remove or replace the nozzle 42. Accordingly, FIGS. 5-9 illustrate a tool 46 for removing or installing the nozzle 42 at the nozzle holder 40. The tool 46 has a generally elongated rod-like body 47 that includes a first end or removal end 48, and an opposite second end or installation end 50. The removal end 48 defines a pocket 52 (FIG. 9) and includes an annular rim 54 communicating with the pocket 52 and having a diameter corresponding to a diameter of the retractable outer ring 44 of the nozzle holder 40. The installation end 50 defines a recess 56 having a size and shape corresponding to the size and shape of a front portion of the nozzle 42. In the illustrated embodiment, the recess 56 has a hexagonal shape corresponding to the hexagonal shape of the front portion of the nozzle 42. The recess 56 receives the front portion of the nozzle 42 (e.g., by nominal slip fit) to temporarily hold the nozzle 42 so that it may be subsequently installed into the nozzle holder 40 using the tool 46.

The tool 46 also includes nozzle storage receptacles 58 formed in a central region of the body 47 between the removal and installation ends 48, 50. The storage receptacles 58 receive additional nozzles 42 for storage and safekeeping. In the illustrated embodiment, the storage receptacles 58 define respective apertures that extend all the way through the body 47 between opposite first and second lateral sides

4

57, 59 of the body 47. As such, nozzle 42 may be inserted into each storage receptacle 58 from the first lateral side 57 or from the second lateral side 59. In other embodiments (not shown), the storage receptacles may instead define recesses that are open to only one of the lateral sides 57, 59 and are closed to the other lateral side 57, 59.

The tool 46 further includes a flexible arm or clip 60 extending from the body 47 that allows the tool 46 to be secured to an article such as a belt, a strap, a pocket, or the like. For example, FIG. 12 illustrates a pair of backpack straps 62 that can be attached to a backpack for carrying the remote reservoir 14 (FIG. 1). The backpack straps 62 include a pocket 64, and the tool 46 is shown secured to the rim of the pocket 64 via the clip 60. In the illustrated embodiment, the clip 60 is integrally formed with the body 47. In other embodiments (not shown), the clip 60 can be a separate component affixed to the body 47.

The tool 46 further includes first and second planer grip surfaces 66, 68 formed at opposite top and bottom sides 70, 72 of the body 47. The grip surfaces 66, 68 are also formed in the central region of the body 47 and located adjacent the storage receptacles 58. The grip surfaces 66, 68 can be grasped by the user to hold and manipulate the tool 46.

FIGS. 10A-10C illustrate a process for removing the nozzle 42 from the nozzle holder 40 using the tool 46. With reference to FIG. 10A, the removal end 48 of the tool 46 is first inserted into the tube 32 through the outlet opening 38. With reference to FIG. 10B, the annular rim 54 is moved over the nozzle 42 and pressed against the outer ring 44, causing the outer ring 44 to retract from the holding position (FIG. 10A) to the release position (FIG. 10B). With the outer ring 44 retracted, the nozzle 42 is released and ejected from the nozzle holder 40 (FIG. 10C). The ejected nozzle 42 lands within the pocket 52. The tool 46 can then be removed from the tube 32 and the nozzle 42 retrieved from within the pocket 52.

FIGS. 11A-11C illustrate a process for installing the nozzle 42 into the nozzle holder 40 using the tool 46. With reference to FIG. 11A, the front portion of the nozzle 42 is inserted into the recess 56 of the installation end 50 of the tool 46, and the installation end 50 is inserted into the tube 32 through the outlet opening 38. With reference to FIG. 11B, the opposite end of the nozzle 42 is then pressed into the nozzle holder 40 until it reaches a secured position at which the outer ring 44 moves to the holding position (FIG. 11B). With the nozzle 42 secured within the nozzle holder 40, the tool 46 can then be removed from the tube 32 (FIG. 11C).

Various features of the disclosure are set forth in the following claims.

What is claimed is:

1. A tool for operating a nozzle holder, the nozzle holder configured to selectively and removably couple a spray nozzle to a spray device, the tool comprising:
 - an elongated body extending between a first end and a second end opposite the first end, the second end defining a recess configured to selectively receive and hold a portion of the spray nozzle; and
 - a nozzle storage receptacle formed in the elongated body between the first end and the second end, the nozzle storage receptacle configured to selectively receive and hold the spray nozzle;
 wherein the nozzle storage receptacle defines an aperture that extends through the elongated body between a first lateral side of the elongated body and a second lateral side of the elongated body opposite the first lateral side.

5

2. The tool of claim 1, wherein the first end defines an annular rim and a pocket that communicates with the annular rim, the annular rim being configured to selectively engage the nozzle holder to release the spray nozzle from the nozzle holder.

3. The tool of claim 1, further comprising a flexible arm extending from the elongated body.

4. The tool of claim 1, further comprising a pair of planar grip surfaces formed on the elongated body between the first end and the second end.

5. The tool of claim 4, wherein the planar grip surfaces are located adjacent the nozzle storage receptacle.

6. The tool of claim 1, wherein the recess includes a hexagonal shape that corresponds to a hexagonal shape of the portion of the spray nozzle.

7. The tool of claim 1, wherein the nozzle storage receptacle is further configured to receive the spray nozzle inserted from the first lateral side, and wherein the nozzle storage receptacle is further configured to receive the spray nozzle inserted from the second lateral side.

8. A tool for operating a nozzle holder, the nozzle holder configured to selectively and removably couple a spray nozzle to a spray device, the tool comprising:

an elongated body extending between a first end and a second end opposite the first end, the second end defining a recess configured to selectively receive and hold a portion of the spray nozzle; and

a nozzle storage receptacle formed in the elongated body between the first end and the second end, the nozzle storage receptacle configured to selectively hold and receive the spray nozzle inserted from a first lateral side of the elongated body and an opposite, second lateral side of the elongated body.

9. The tool of claim 8, wherein the nozzle storage receptacle is a first nozzle storage receptacle, and the tool further includes a second nozzle storage receptacle formed in the elongated body between the first end and the second end and configured to selectively receive and hold the spray nozzle.

10. The tool of claim 8, wherein the first end is configured to engage the nozzle holder to release spray nozzle.

11. The tool of claim 10, wherein the first end defines a pocket that does not communicate with the recess, and wherein the pocket is configured to receive the spray nozzle.

6

12. The tool of claim 11, wherein the first end includes an annular rim that communicates with the pocket.

13. The tool of claim 8, wherein the recess includes a hexagonal shape that corresponds to a hexagonal shape of the portion of the spray nozzle.

14. The tool of claim 8, further comprising a pair of planar grip surfaces formed on the elongated body between the first end and the second end, wherein the planar grip surfaces are located adjacent the nozzle storage receptacle.

15. A tool for operating a nozzle holder, the nozzle holder configured to selectively and removably couple a spray nozzle to a spray device, the tool comprising:

an elongated body extending between a first end and a second end;

an annular rim defined by the first end, the annular rim being configured to selectively engage the nozzle holder to release the spray nozzle; and

a nozzle storage receptacle formed in the elongated body between the first end and the second end, the nozzle storage receptacle configured to selectively hold and receive the spray nozzle inserted from a first lateral side of the elongated body and an opposite, second lateral side of the elongated body.

16. The tool of claim 15, wherein the nozzle storage receptacle is a first nozzle storage receptacle, and the tool further includes a second nozzle storage receptacle formed in the elongated body between the first end and the second end and configured to selectively receive and hold the spray nozzle.

17. The tool of claim 15, wherein the first end defines a pocket that communicates with the annular rim, the pocket being configured to receive the spray nozzle.

18. The tool of claim 15, wherein the second end defines a recess configured to selectively receive and hold a portion of the spray nozzle.

19. The tool of claim 18, wherein the recess includes a hexagonal shape that corresponds to a hexagonal shape of the portion of the spray nozzle.

20. The tool of claim 15, further comprising a pair of planar grip surfaces formed on the elongated body between the first end and the second end, wherein the planar grip surfaces are located adjacent the nozzle storage receptacle.

* * * * *